INCH-POUND

MIL-DTL-85097/3D(OS)
20 March 2001
SUPERCEDING
MIL-A-85097/3C(OS)
5 December 1994

MILITARY ASSOCIATED DETAIL SPECIFICATION

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) ROCKET MOTOR (LOADED ASSEMBLY) MARK 79 MODS 1 & 2

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Mark 79 Mods 1 & 2 Rocket Motor described herein shall consist of this specification and the latest issue of MIL-DTL-85097.

1. SCOPE

1.1 Scope. This document covers requirements unique to the Rocket Motor Mark 79 Mods 1 & 2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-DTL-85097 Aircrew Escape Propulsion Systems (AEPS), General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Documentation Branch (Code 4230), Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

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(Unless otherwise indicated, copies of Department of Defense specifications and standards are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications are in addition to those listed in the basic specification, MIL-DTL-85097, and form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2)

SPECIFICATIONS

NAVAL SURFACE WARFARE CENTER (CAGE Code 14083)

510-174TD-004	Propellant, NSWC/IH-EC-8
511-174TD-004	Propellant, NOSIH-BC-11, NSWC/IH-EC-02, NSWC/IH-EC-04, and NSWC/IH-EC-05

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

DL672AS200	Rocket Motor Assembly Mk 79 Mods 1 & 2
672AS200	Rocket Motor Assembly Mk 79 Mods 1 & 2
672AS231	Motor Subassembly

(Application for copies of specifications and drawings should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Code 8410P, 101 Strauss Avenue, Indian Head, MD 20640-5035.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements in this specification, unless in direct conflict with the basic specification, MIL-DTL-85097, shall be considered additional, not substitute. The Mark 79 Mods 1 & 2 Rocket Motor shall conform to DL672AS200. If there is a direct conflict with the basic specification, this document shall take precedence.

3.2 Propellant

3.2.1 Mk 79 Mod 1. The Mk 79 Mod 1 propellant shall conform to the requirements of MIL-DTL-85097 and 511-174TD-004. Testing shall be in accordance with 4.1.1.1.

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3.2.2 Mk 79 Mod 2. The Mk 79 Mod 2 propellant shall conform to the requirements of MIL-DTL-85097 and 510-174TD-004. Testing shall be in accordance with 4.1.1.2.

3.3 Propellant aging.

3.3.1 Mk 79 Mod 1. The Mk 79 Mod 1 propellant shall meet the aging requirements of MIL-DTL-85097 and 511-174TD-004. Testing shall be in accordance with 4.1.1.1.

3.3.2 Mk 79 Mod 2. The Mk 79 Mod 2 propellant shall meet the aging requirements of MIL-DTL-85097 and 510-174TD-004. Testing shall be in accordance with 4.1.1.2.

3.4 Radiographic inspection. The Mk 79 Mods 1 & 2 Rocket Motor shall have no defects when subjected to radiographic inspection in accordance with 4.1.2.

2.5 Ballistic performance. The ballistic performance shall conform to table I. Testing shall be conducted in accordance with 4.1.3.

TABLE I. Ballistic performance requirements.

Characteristic	Value		Temperature (±5°F)
	Min	Max	
Total impulse (lb-sec)	580	640	-40, 160
Action time (ms)	190	320	-40, 160
Thrust (lb)	-	3800	-40, 160
Ignition delay (ms)	-	25	-40, 160
Initiation pressure (psig)	400	650	-40, 160
Time to shear (ms)	-	10	-40, 160
Rise time (ms) (see 6.2.1)	-	30	-40, 160

4. VERIFICATION.

4.1 First article/quality conformance. First article and quality conformance tests shall be performed as specified in MIL-DTL-85097 and shall consist of the inspections and tests listed in 4.1.1 through 4.1.3.

4.1.1 Propellant and propellant aging.

4.1.1.1 Mk 79 Mod 1. The contractor shall submit MK 79 Mod 1 propellant and propellant aging test results in accordance with MIL-DTL-85097 and 511-174TD-004. Units shall meet the requirements of 3.2.1 and 3.3.1.

4.1.1.2 Mk 79 Mod 2. The contractor shall submit MK 79 Mod 2 propellant and propellant aging test results in accordance with MIL-DTL-85097 and 510-174TD-004. Units shall meet the requirements of 3.2.2 and 3.3.2.

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4.1.2 Radiographic inspection. Units shall meet the requirements of 3.4 when subjected to radiographic inspection. Radiographic inspection shall be as follows:

a. Inspect the propellant grain assembly at the head and aft end. Four longitudinal views shall be taken at each end, 45° apart, as specified by drawing 672AS231.

b. Inspect the rocket motor assembly in two longitudinal planes 90° apart, as specified by drawing 672AS200.

4.1.3 Ballistic performance. Ballistic testing of the Mk 79 Mods 1 & 2 Rocket Motor shall be performed in accordance with MIL-DTL-85097. Units shall meet the requirements of 3.5.

4.1.3.1 Conditioning temperatures. The conditioning temperatures and tolerance shall be as follows:

a. Cold: –40 ± 5°F
b. Hot: +160 ± 5°F

4.1.3.2 Conditioning period. Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: Not less than 4 hours
b. Temperature cycling: Not less than 5 hours

5. PACKAGING

(This section is not applicable to this specification.)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Mk 79 Mods 1 & 2 Seatback Rocket Motor assists in ejecting personnel from the T/AV-8B and A-7 aircraft during an emergency. The rocket motor was developed for use in only those military aircraft, and therefore has no commercial application.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).

c. That the safety precaution requirements of the “Contractors’ Safety Manual for Ammunition, Explosives and Related Dangerous Material” (DOD 4145.26M) are applicable and should be specified in the contract as required by the Federal Acquisition Regulations (FAR) 2.3.3.

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NOTE

When this document is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.3 Definition.

6.3.1 Rise time. The interval from 10% to 75% of the maximum thrust during the initial rise on the thrust curve occurring just after ignition.

6.4 Subject term (keyword) listing.

Seatback
T/AV-8B
A-7

6.5 Changes from previous issue. Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing Activity:
Navy - OS
(Project 1377-0151)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-85097/3D	2. DOCUMENT DATE (YYMMDD) 010320

3. DOCUMENT TITLE
AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) ROCKET MOTOR (LOADED ASSEMBLY) MARK 79 MODS 1 & 2

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial: 301-744-4700 (2) DSN: 354-4700
c. ADDRESS (*Include Zip Code*) Documentation Branch (Code 4230) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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